

Diesel Engines**12V 2000 G****16V 2000 G****Stationary Power Generation:
Continuous Operation****Maintenance Schedule****M050517/05E**

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean,
- use only dry, clean compressed air,
- use only clean, filtered raw water.

Moreover, it is essential that:

- maintenance tasks be completed by trained personnel,
- suitable tools be used,
- genuine spare parts and fluids and lubricants as per the current MTU Fluids and Lubricants Specification be used.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Follow the manufacturer's maintenance instructions for components belonging to the power plant but not listed in this Maintenance Schedule.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept and features the maintenance echelons W1 to W6 as outlined below.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

Main features of the maintenance echelons:

- W1: Operational checks
- W2, W3 and W4: Periodic maintenance tasks to be performed during out-of-service periods without the need for engine disassembly.
- W5: This echelon requires partial engine disassembly.
- W6: Major overhaul. This echelon requires complete engine disassembly.

The Maintenance Schedule is based on the agreed power requirements.

The time intervals according to which the W2 to W6 Maintenance Echelons and the relevant checks and tasks involved are to be completed are based on operational experience. They have been determined so as to ensure correct engine operation until the next scheduled maintenance echelon.

Specific operational conditions may require modification of the Maintenance Schedule.

Note:

The codes on the following pages refer to the section numbers in Section G of the Engine Operation Manual.

This Maintenance Schedule may refer to parts which are not installed on your engine, these may be disregarded.

Application Group

3A: Continuous power

3B: Prime power

Maintenance Frequency Chart

		3A	3B
Maintenance Echelon	W1: Operational checks, daily	X	X
	W2: Operating hours	500	500
	Limit, months	6	6
	W3: Operating hours	1 000	1 000
	Limit, year	1	1
	W4: Operating hours	4 000	3 000
	Limit, year(s)	4	4
	W5: Operating hours	12 000	9 000
	Limit, year(s)	8	8
	W6: See page 3		
	Maintenance Echelon W6		

Maintenance Task Schedule

Code No.	One-time operations after the first 50 operating hours. – With a new engine, after W5 maintenance echelon, or after W6 major overhaul	
G055.050.01	Valve gear	Check valve clearances
G083.101.09	Fuel pre-filter	Check, replace filter inserts if necessary
G140.000.03	Exhaust system	Check tightness of securing screws and nuts
G202.051.01	Engine coolant pump	Check relief bore for obstructions
G207.051.01	Raw water pump	Check relief bore for obstructions
G231.051.01	Engine mounts	Check tightness of securing screws and nuts

Code No.	Maintenance Echelon W1: – Operational checks	
G000.000.13	Engine operation	Check speed
G000.000.15	Engine operation	Check pressures (if gauges are provided)
G000.000.17	Engine operation	Check temperatures (if gauges are provided)
G000.000.19	Engine operation	Check running noises
G000.000.21	Engine operation	Check engine and external pipework for leaks,
G000.000.37	Engine operation	Test run at not below 1/3 load and at least until steady-state temperature is reached
G121.052.01	Air filter	Check contamination indicator
G140.000.05	Exhaust system	Check exhaust gas colour
G180.000.01	Engine oil	Check level
G202.000.03	Engine coolant	Check level
G213.051.01	Charging generator drive	Check belt condition and tension
G221.051.01	Engine cooling fan	Check belt condition and tension